

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Led design of a multi-agent orchestration system (LangGraph, OpenAI Agents SDK, MCP tool-calling) with RAG retrieval over Pinecone, serving millions of daily transactions at 99%+ uptime and sub-200ms latency.
- Built real-time, SSE-based streaming layer surfacing live agent output to end users; owned Python/FastAPI backend services end-to-end across microservices handling high-concurrency production traffic.
- Architected AWS serverless infrastructure (Lambda, S3, DynamoDB, API Gateway, SageMaker) exposing production APIs at sub-200ms latency, supporting high-throughput, data-intensive workloads at scale.

QureAI

Jan 2024 – May 2025

ML Infrastructure Engineer

- Drove a RAG retrieval redesign improving recall 61% to 89% on a 2K-query eval set via hybrid dense-sparse retrieval and re-ranking, using Redis-backed checkpointing for fault-tolerant stateful recovery.
- Led migration of synchronous model serving to async vLLM on GCP Cloud Run with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java microservices exposing REST APIs for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS) in a distributed, fault-tolerant architecture.
- Drove a redesign of a nightly batch data pipeline aggregating financial transactions, cutting downstream-blocking processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams, reducing production escalations 60% in one quarter through code reviews and design write-ups.

Projects

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Full-stack production platform with rubric-based scoring, drift detection, and per-call latency tracking across model versions; Python/FastAPI backend, React/TypeScript dashboard, PostgreSQL.

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL

Backend & Real-Time: FastAPI, Flask, Node.js, REST APIs, Microservices, SSE (real-time streaming), Redis, PostgreSQL

GenAI & Agents: LangGraph, LangChain, OpenAI Agents SDK, MCP, RAG, Vector DBs (Pinecone, FAISS)

Web & Frontend: React, TypeScript, HTML/CSS

Tools & Cloud: Git, Docker, Kubernetes, CI/CD, AWS (Lambda, EC2, S3), GCP (Cloud Run)

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer